

Quiz (Habanero)

Q1. What is the GCF of 12 and 18 using factors?

- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 12

Q2. Find the LCM of 9 and 15 using multiples.

- (A) 15
- (B) 30
- (C) 45
- (D) 60
- (E) 90

Q3. A storm is forecasted to bring 24 inches of rain to a coastal town. If the town's water reservoir can hold 48 inches of water, what is the GCF of the total rainfall and the reservoir's capacity?

- (A) 1
- (B) 2
- (C) 4
- (D) 8
- (E) 12

Q4. The temperature in a city changes by 8 degrees every hour. If it also changes by 12 degrees every two hours, what is the LCM of these two temperature changes?

- (A) 8
- (B) 12
- (C) 16
- (D) 24
- (E) 32

Q5. A group of scientists are studying the population growth of a certain ecosystem. They find that the population of one

species increases by 15 every 3 days, and another species increases by 20 every 5 days. What is the LCM of the time intervals for these population increases?

- (A) 3
- (B) 5
- (C) 10
- (D) 15
- (E) 15

Q6. What is the GCF of 24, 36, and 48 using prime factorization?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 12

Q7. The rainfall in a region follows a cycle of 18 days of rain and 12 days of dry weather. What is the LCM of the days of rain and dry weather?

- (A) 12
- (B) 18
- (C) 30
- (D) 36
- (E) 54

Q8. A team of environmentalists are monitoring the water level of a lake. The water level rises by 9 inches every 2 hours and falls by 6 inches every 3 hours. What is the GCF of the time intervals for these changes?

- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 9

Open Ended Questions (FRQ)

Q9. A certain type of plant grows 8 inches every 4 days and another type grows 12 inches every 6 days. Find the LCM of the time intervals and the GCF of the growth rates. Show your work and explain your reasoning.

Q10. A meteorologist is predicting the temperature changes in a region. The temperature changes by 15 degrees every 3 hours and by 20 degrees every 5 hours. Find the LCM of the time intervals and the GCF of the temperature changes. Provide a real-world application of your answer.

Chef's Tips

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- Tip 1: Ensure students understand the concepts of GCF and LCM before applying them to real-world problems.
- Tip 2: Remind students to show their work and explain their reasoning for free-response questions.
- Tip 3: Encourage students to use visual aids or diagrams to help solve problems.